

Supplement 27.2 — $|X \setminus \{a\}| = |X| - 1$

CS 173: Discrete Structures · Functions & Cardinality

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1 Context

Remark (Theorem (!)). For any finite sets X and Y , if $Y \subseteq X$, then

$$|X \setminus Y| = |X| - |Y|.$$

The theorem below is the special case $Y = \{a\}$, which is proved first and used to establish the general result.

Theorem 1.1 (Theorem (?)). For any finite set X and any a , if $a \in X$, then

$$|X \setminus \{a\}| = |X| - 1.$$

2 Proof

Proof. Let X be a finite set, let a be a set, and assume $a \in X$.

Since X is finite, there exists $n \in \mathbb{N}$ such that $|X| = |n|$. (Recall $n = \{0, 1, 2, \dots, n-1\}$.) This means there exists a bijection $f: X \rightarrow n$; in particular, f is injective and surjective.

Define $g: X \setminus \{a\} \rightarrow n-1$ by: for any $b \in X \setminus \{a\}$,

$$g(b) := \begin{cases} f(b) & \text{if } f(b) < f(a), \\ f(b) - 1 & \text{if } f(b) > f(a). \end{cases}$$

Remark. Since f is a bijection and $b \neq a$, we have $f(b) \neq f(a)$, so $f(b) < f(a)$ or $f(b) > f(a)$ — the two cases are exhaustive and mutually exclusive, so g is well-defined.

Part 1: Injectivity. We prove $(\forall b_1, b_2 \in X \setminus \{a\})(g(b_1) = g(b_2) \Rightarrow b_1 = b_2)$.

Let $b_1, b_2 \in X \setminus \{a\}$ and assume $g(b_1) = g(b_2)$.

Case 1: $f(b_1) < f(a)$.

By definition of g , $g(b_1) = f(b_1)$, so $f(b_1) = g(b_1) = g(b_2)$.

Towards a contradiction, suppose $f(b_2) > f(a)$. Then by definition, $g(b_2) = f(b_2) - 1 \geq f(a)$ since $f(b_2) > f(a)$. So $f(b_1) = g(b_1) = g(b_2) \geq f(a)$, contradicting $f(b_1) < f(a)$. $\Rightarrow \Leftarrow$

Therefore $f(b_2) < f(a)$, so by definition $g(b_2) = f(b_2)$. Thus $f(b_1) = f(b_2)$, so $b_1 = b_2$ because f is injective.

Case 2: $f(b_1) > f(a)$.

By definition of g , $g(b_1) = f(b_1) - 1$, so $f(b_1) - 1 = g(b_1) = g(b_2)$.

Towards a contradiction, suppose $f(b_2) < f(a)$. Then by definition, $g(b_2) = f(b_2)$, so $f(b_1) - 1 = f(b_2)$. Since $f(b_1) > f(a)$ we have $f(b_1) \geq f(a) + 1$, hence $f(b_2) = f(b_1) - 1 \geq f(a)$, contradicting $f(b_2) < f(a)$. $\Rightarrow \Leftarrow$

Therefore $f(b_2) > f(a)$, so by definition $g(b_2) = f(b_2) - 1$. Thus $f(b_1) - 1 = f(b_2) - 1$, implying $f(b_1) = f(b_2)$, so $\boxed{b_1 = b_2}$ because f is injective.

In both cases $g(b_1) = g(b_2) \Rightarrow b_1 = b_2$, so g is injective.

Part 2: Surjectivity. We prove $(\forall y \in n-1)(\exists b \in X \setminus \{a\})(g(b) = y)$.

Let $y \in n-1$. We know $n-1 < n$, so $n-1 \in n$, so $n-1 \subseteq n$, which gives $y \in n$.

Case 1: $y < f(a)$.

Since f is surjective and $y \in n$, there exists $b \in X$ such that $f(b) = y$.

We know $b \neq a$ because if $b = a$, then $y = f(b) = f(a) < f(a)$, a contradiction. $\Rightarrow \Leftarrow$

So $b \in X \setminus \{a\}$ by definition. Since $f(b) = y < f(a)$, by definition of g :

$$g(b) = f(b) = y.$$

Case 2: $y \geq f(a)$.

Notice $y+1 > y \geq f(a)$, so $y+1 > f(a)$. Since $y \in n-1$, we know $y < n-1$, so $y+1 < n$, hence $y+1 \in n$.

So, since f is surjective, there exists $b \in X$ such that $f(b) = y+1$.

We know $b \neq a$ because if $b = a$, then $f(a) = f(b) = y+1 > f(a)$, a contradiction. $\Rightarrow \Leftarrow$

So $b \in X \setminus \{a\}$. Since $f(b) = y+1 > f(a)$, by definition of g :

$$g(b) = f(b) - 1 = (y+1) - 1 = y.$$

In both cases we found $b \in X \setminus \{a\}$ with $g(b) = y$, so g is surjective.

So, since g is injective and surjective, g is a bijection, and therefore $|X \setminus \{a\}| = n-1$ by definition. $\square$